

Power-efficient Planar Quadrotor Control through Convex Optimization

I. MOTIVATION

Unmanned air vehicles (UAVs) are ubiquitous in real life. It has been widely used in fields like delivery, rescue, and performing many dangerous tasks that humans are incapable of. However, it is inefficient to control it manually or with simple control algorithms due to the amount of control required and the limited power available. In this project, our goal is to minimize the time it takes a UAV to complete a simulated flight task between obstacles. To be more specific, we will use the planar quadrotor model mentioned in the lecture and build a 2D environment with borders and gate obstacles for the simulation.

II. TECHNICAL APPROACH

Milestone I

Mathematical Model Establishment: Following the lecture notes, the planar quadrotor dynamics are governed by:

$$m\ddot{q}_1(t) = -\sin(q_3(t))(u_1(t) + u_2(t)) \quad (1)$$

$$m\ddot{q}_2(t) + mg = \cos(q_3(t))(u_1(t) + u_2(t)) \quad (2)$$

$$I\ddot{q}_3(t) = r(u_2(t) - u_1(t)) \quad (3)$$

To enable trajectory planning, the horizontal and vertical positions are parameterized as degree-7 polynomials in time:

$$q_1 = a_7t^7 + a_6t^6 + a_5t^5 + a_4t^4 + a_3t^3 + a_2t^2 + a_1t + a_0$$

$$q_2 = b_7t^7 + b_6t^6 + b_5t^5 + b_4t^4 + b_3t^3 + b_2t^2 + b_1t + b_0$$

Convex Optimization for Quadrotor control: We first consider a simple problem, where the quadrotor travels between two positions subject only to a maximum thrust limit and the physical properties. The optimization is formulated as:

$$\min T_{total} = \sum_k h \quad (4)$$

$$\text{s.t. } \mathbf{q} \text{ satisfies dynamics (1)–(3)} \quad (5)$$

$$\mathbf{u} \leq \mathbf{U}_{\max} \quad (6)$$

Where h denotes the discrete time step size, and k is the fixed number of steps in the current task.

Milestone II

Environment Setup for Non-convex Task: We extend the problem to a 2D racing scenario, in which the quadrotor departs from an initial position, passes through a sequence of gate checkpoints in order, and arrives at a designated destination.

Task Decomposition: The set becomes non-convex due to the constraints of collision avoidance and thrust bound.

$$\begin{aligned} \min & T_{total} \\ \text{s.t.} & \text{ constraints from Eq. 5 and 6} \\ & (x_k, y_k) \notin \mathbb{X}, \quad \forall k = 1, \dots, n \end{aligned}$$

where (x_k, y_k) denotes the k -th sampled coordinate along the drone's contour, and $\mathbb{X}$ is the set of coordinates occupied by gate borders. This constraint renders the feasible set non-convex.

To turn it into a convex set, this project decomposes the full trajectory into segments, each connecting two consecutive sides of a gate. The overall time minimization problem then reduces to independently minimizing the flight time of each segment T_m in n sub-tasks. Hence, the optimization problem becomes:

$$\min T_m = h_m k_m \quad (7)$$

$$\text{s.t. constraints from Eq. 5 and 6}$$

$$\mathbf{q}_m = \mathbf{q}_{m-1} \quad (8)$$

Where $T_{total} = \sum_{m=1}^n T_m$, Eq. 8 guarantees that the initial state of the current sub-task in continuous with the final state of the previous sub-task.

To solve this convex optimization problem, we'll use the `cvxpy` package as the problem solver. The optimized variables are $\mathbf{a}$, $\mathbf{b}$ (params in the smooth functions), and h_m .

Milestone III

We will plot the quadrotor's trajectory as well as the visualization to validate our design and. If time permits, we'll figure out if we can let the algorithm optimize the path instead of specifying the final states in each sub-task.

III. EXPECTED OUTCOME

We aim to develop a complete planar quadrotor control pipeline-grounded in the theory covered in lecture, and demonstrate its effectiveness in the simulated environment by converting a non-convex problem into a convex optimization formulation. The solution should shows a elaborated and decent control trajectory.

IV. TEAM INFORMATION

We'll collaborate on the GitHub repository provided above. Besides, this project will be based on two students' collaboration: Yuchen Chen yuchen_chen@ucsb.edu and Tianyi Chen tianyi_chen@ucsb.edu.

⁰Project Under UCSB ECE594T in Spring 2026; Code available at: <https://github.com/apprentice922/UCSB-ECE594T-robotics-control>